

BREAKING THE SOUND BARRIER



STOOP TO ENTER

Chuck Yeager demonstrates how to enter the cockpit of the Bell X-1 Glamorous Glennis. For experimental flights, the pilot left the ground as a passenger on the B-29 or B-50 mother ship and climbed down to the X-1 cockpit as the bomber gained altitude.

MOTHER SHIP READY

On October 14, 1947, the day that Yeager broke the sound barrier, the B-29 mother ship stands ready to take off, with the Bell X-1 mounted half inside its bomb bay. Experimental flights always took place early in the morning before the desert air heated up.



MOMENT OF TRUTH

The Bell X-1 is released from its mother ship at above 20,000ft (6,000m). The X-1 pilot then put down the nose to gather speed in a gliding dive before igniting one or more rocket chambers. Yeager described this as "the moment of truth: if you are gonna be blown up, this is likely to be when."

ON OCTOBER 14, 1947, a sound never before heard on earth echoed over the dry lake beds of the Mojave desert, California: the sonic boom from an aircraft reaching Mach 1. The aircraft was the Bell X-1 and the man at the controls was a fighter pilot from West Virginia, Chuck Yeager.

Yeager had nearly not made it into the air that morning. Two days earlier, he had ridden a horse into a gate and cracked two ribs. But there was no way the West Virginia pilot was going to pull out of a flight planned to test the sound barrier. Yeager's injury meant that climbing down the ladder into the X-1 cockpit, as it hung in the bomb bay of its B-29 mother plane at 7,000ft (2,000m), was agony. And it meant adding to Yeager's normal flight equipment a piece of broomstick, to use as a

lever to help him lock the cockpit door from the inside – an inspired piece of improvisation made up just before takeoff.

When the B-29 had carried the X-1 up to 20,000ft (6,000m), the pilot asked Yeager whether he was ready to go. "Hell, yes," came the West Virginia drawl, "Let's get it over with." As was by then routine – it was Yeager's ninth powered X-1 flight, gradually pushing up the Mach numbers – the B-29 went into a dive and released the rocket plane like a gliding bomb. Yeager graphically described the experience of being air-launched in the X-1: "The bomb shackle release jolts you up from your seat, and as you sail out of the dark bomb bay the sun explodes in brightness."

Yeager put the nose down to gather more speed and keep from stalling, then leveled out and ignited all four rocket chambers in quick succession. Slammed back in his seat, he pointed the nose up and accelerated into the upper atmosphere.

At .88 Mach, the X-1 began to shake. This was an experience Yeager had run into before, the encounter with shockwaves that would be followed